

StitchPerfect

Revolutionizing Tailor Shops

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Digitize Orders Precision Measurements Faster Deliveries Data-Driven Growth

Problem Statement Title

Enhancing Tailor Shop Efficiency through Digital Management and Automation.

1 Higher-order Goal

Local tailor shops largely depend on manual, paper-based processes for taking measurements, tracking orders, and coordinating with staff. This leads to measurement errors, long turnaround times, and missed business opportunities. StitchPerfect aims to digitize order intake, standardize measurement storage, and provide real-time visibility into shop operations for both staff and customers, reducing rework and improving delivery speed.

2 Problem Statement

People running or working in local tailor shops routinely face avoidable inefficiencies caused by manual record-keeping:

- **Measurement mistakes:** measurements recorded on paper are easy to misplace, misread, or lose between visits.
- **Long turnarounds:** lack of a structured order pipeline makes it difficult to track where a garment is in the stitching process.
- **Lost sales:** without visibility into inventory and order status, shops struggle to manage repeat customers and growth.
- **No customer visibility:** customers have no way to track their order or verify progress before collection.
- **Fragmented staff coordination:** new staff onboarding is slow, and task assignment across workers is informal.

Why this matters: A digital, tailor-first platform that standardizes measurements and automates the order workflow can materially reduce errors and delivery time while giving shop owners the operational visibility needed to scale.

3 Proposed Solution

3.1 Overview

StitchPerfect is a digital management platform built specifically for tailor shops and boutiques. Rather than being a generic order-tracking tool, it combines measurement digitization, media-based proof-of-work, inventory tracking, and customer communication into a single tailor-first system.

The proposed solution consists of the following components:

- **Order Management & Workflow:** digital order intake, task cards, and status tracking through the full stitching pipeline.
- **Media Management (Clouinary):** photo-based proof-of-work, style references, and design previews stored in the cloud.
- **Inventory & Material Tracking:** real-time fabric and material stock control linked to active orders.
- **Customer Communication & Payments:** automated updates, reminders, and integrated payment collection.
- **Measurement Explorer:** centralized, searchable storage of customer measurement profiles for faster reorders.
- **Worker Console:** task queues and status updates for stitching staff.
- **Admin Panel:** full operational oversight of orders, workers, and delivery status, with support for multi-shop, role-based access.

3.2 How this solves the problem

By digitizing measurements, automating the order workflow, and providing customer-facing tracking with backend analytics, StitchPerfect reduces rework caused by lost or miscommunicated measurements, shortens lead times through clearer task ownership, and increases shop revenue by enabling faster repeat orders and better inventory decisions.

3.3 Who benefits

- **Tailors & stitching staff:** simpler daily queues, clear task cards, fewer corrections, saved measurement profiles, faster onboarding.
- **Shops & owners:** visibility into all orders, cost and profitability per garment, inventory control, supplier integration, scalability to multiple outlets.
- **Customers:** order tracking, preview images, fewer fittings, trusted proof-of-work, faster delivery and easier reorders.
- **Suppliers & vendors:** electronic purchase orders, lead-time tracking, demand forecasts from analytics.
- **Investors / business partners:** monetization metrics, growth projections, low customer acquisition cost via shop partnerships.

4 System Context Diagram

Figure 1 provides a high-level view of StitchPerfect and its interaction with the primary external actors: customers, tailoring staff, shop owners/administrators, and suppliers. This places the StitchPerfect platform at the center, unifying the tailoring value chain into a single digital system.

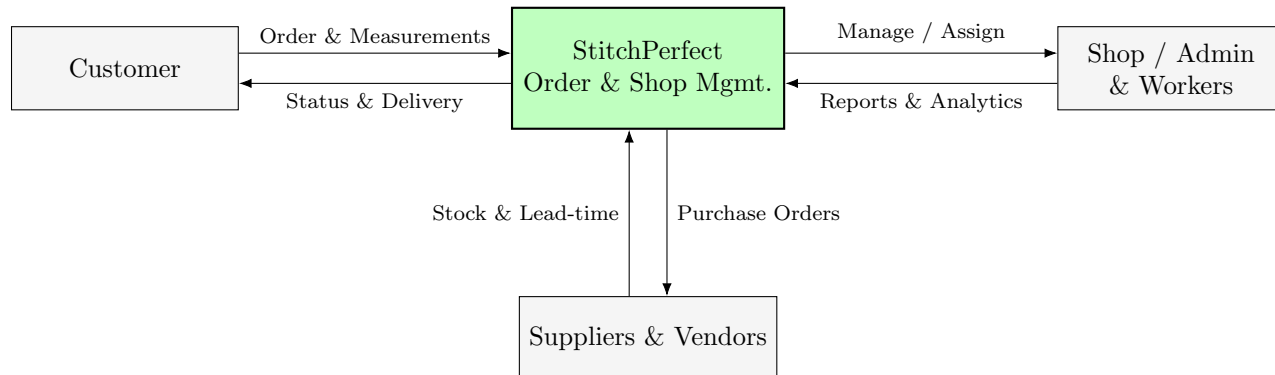


Figure 1: Level-0 Context Diagram for StitchPerfect

5 Use Case Diagram

Figure 2 illustrates the major functional interactions between customers, tailoring staff/workers, and shop administrators.

6 Level-1 Data Flow Diagram

Figure 3 expands the system context into the main functional processes and data stores within StitchPerfect. Process 6.0 (Customer Comm. & Payments) is also responsible for pushing status updates and reminders back out to the customer, alongside the direct order and measurement intake shown at process 1.0.

7 Solution Approach

7.1 Implementation flow

The administrator signs in to a central Admin Panel that branches into the Worker Console and Measurement Explorer, and further into Order Delivery, Ready Products, and the Worker List, giving full operational control over daily shop activity.

7.2 Technology stack

StitchPerfect is built on a lightweight, scalable stack:

- **Bootstrap** – responsive front-end UI.
- **Node.js** – backend server.
- **jQuery** – dynamic front-end interactivity.

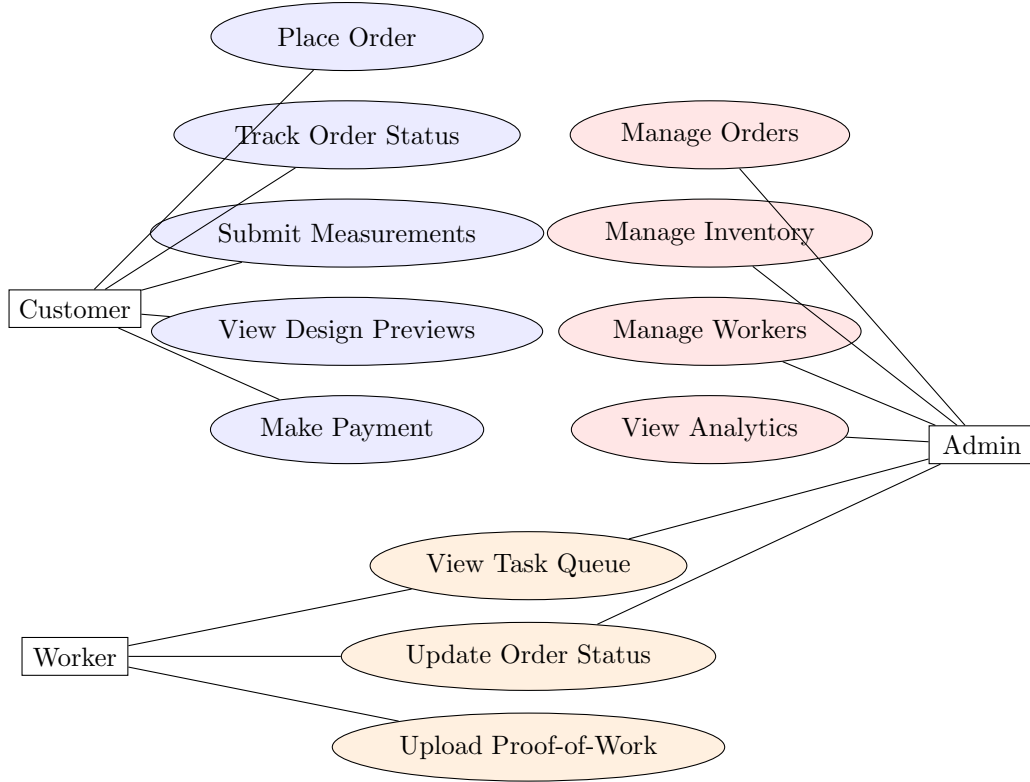


Figure 2: Use Case Diagram for StitchPerfect

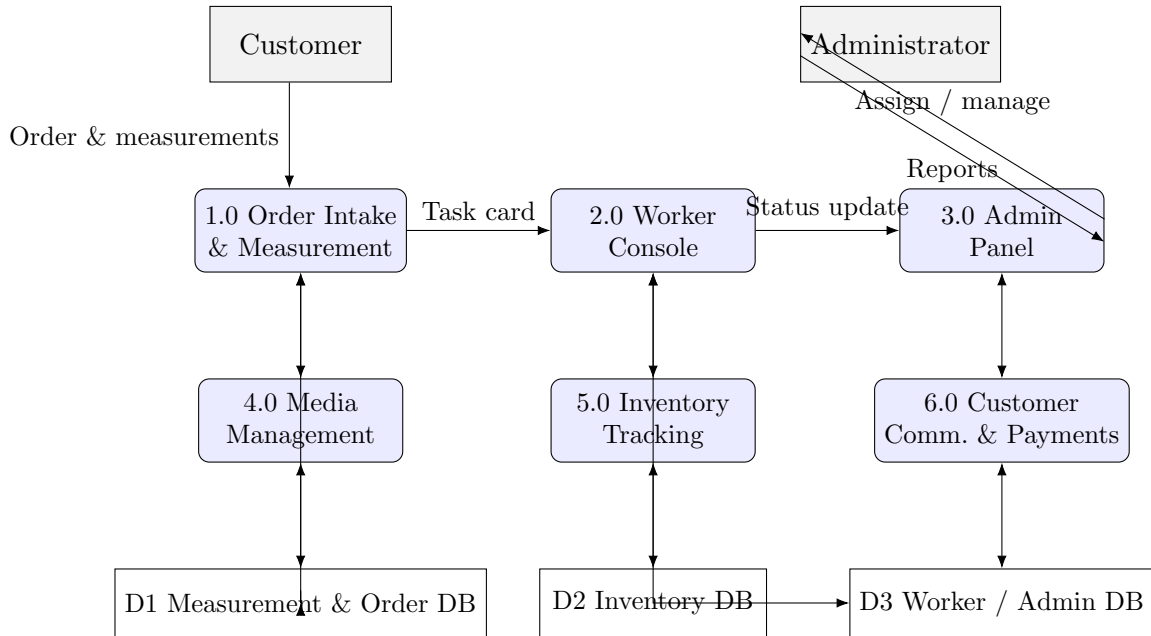


Figure 3: Level-1 Data Flow Diagram for StitchPerfect

- **Cloudinary** – media and image management (proof-of-work, style references).
- **Aiven** – managed cloud database infrastructure.

7.3 Database design

The schema centers on the `mesurement` table, which records every order and links back to `customerlist` (via mobile number) and `workerlist` (via worker name) – a one-to-many relationship in both cases, since a single customer can place many orders and a single worker can be assigned to many orders. The `pass` table stores admin login credentials independently of the order data.

Table	Key
<code>customerlist</code>	PK: <code>mobile</code>
<code>workerlist</code>	PK: <code>wname</code>
<code>mesurement</code>	PK: <code>orderid</code> ; FK: <code>mobile</code> → <code>customerlist.mobile</code> ; FK: <code>worker</code> → <code>workerlist.wname</code>
<code>pass</code>	PK: <code>id</code> (standalone, no foreign keys)

7.4 Ecosystem

The StitchPerfect platform sits at the center of the tailoring ecosystem, connecting tailoring staff, shop owners, and customers on one side with suppliers, investors, and multi-shop operators on the other – unifying the entire tailoring value chain into a single digital system, and supporting multi-shop management with role-based access as the platform scales.